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# Albedo factor determination of some selected 3d alloy samples

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ABSTRACT

The concept of albedo is used to describe the reflected incident radiation from a surface. Actually, this reflected radiation consists of two-component that enter the target samples and then backscattered from the target surfaces. The information about these components of the albedo phenomenon serves the solution of the albedo factor parameters that namely, the albedo number, energy, and dose. The albedo parameters of scientific importance 3d alloy samples have been measured and calculated, in this present work. The calculated albedo parameters plotted as a function of the mean atomic number. Eventually, this work shows that the albedo factor parameters of selected samples decrease with the increasing mean atomic number.

## 1. Introduction

X-ray absorption and emission processes are used in many analytical techniques in the study of a wide variety of materials. Recently, in energy dispersive X-ray fluorescence (EDXRF) analysis; this method has increased due to the simultaneous multielement measurement capability and simple estimation of the characteristic X-ray emission and absorption regardless of the physical and chemical conditions of the elements in the sample. Besides, the EDXRF technique is used in the measurement of the radiation shielding parameters of technological importance elements and different compounds. The radiation shielding parameters are usually determined by the backscattering proportion of incident gamma or x-ray photons emitted from the source. The EDXRF analysis provides us with determining the backscattered photon ratios from the target materials. The photon backscattering intensities are related to the chemical composition of the target materials and the conditions of the experimental setup (AKKUŞ and YILMAZ, 2019).

The backscattering intensities of any target materials are used in the determination procedure of the albedo factor. The albedo factor is known as the reflection capability of the selected material. The albedo factor feature of any material is different from each other and effect from the material thicknesses, the atomic composition, the atomic or effective atomic number, and the energy of the incident electromagnetic radiation. The albedo factor parameter information is used in the radiation shielding material manufacture industries.

The gamma-ray shielding features and albedo factor determination of different scientific importance materials has been attracted to many researchers in recent years. They have used different scattering angles, material thicknesses, experimental setups, and incident gamma sources. Akman et al. (AKMAN et al., 2019a) have investigated the photon shielding performance of some selected alloys (Ag92·5/Cu7.5, Ag72/Cu28, Pd94/Cr6 and Pd60/Cu40). The shielding performance of the alloy samples was tested in 81–1333 keV energy range. The experimental calculated values were compared with theoretical values that obtained from WinXCOM and MCNPX code at all energies. Agar et al. (AGAR et al., 2019) have measured the photon interaction properties of Pb and Ag containing alloys. Kaur et al. (KAUR et al., 2020) have measured the effective atomic number and the albedo factor properties of $Pb_{60}Sn_{20}Zn_xCd_{(20-x)}$ type alloys and some metallic samples. The alloy samples were prepared by using the melt quench technique and the backscattered photon intensities were determined. They reported that the albedo factors of selected samples were decreased with an increased atomic number in the dominant region of the Compton Scattering region. Besides, the albedo factors were decreased with the increasing incident photon energies. Akkuş and Yılmaz have measured 14 different samples at 180° scattering angle. The mean atomic number of the target samples was varied at $9.743 \leq Z \leq 83.00$ scales. The target samples were irradiated by 662 keV gamma photons that were emitted from $^{137}Cs$ radioactive source. They reported that the albedo factor of selected samples was a decrease by the mean atomic number. Also, the high correlation between the mean atomic number and the albedo factor reported (AKKUŞ and YILMAZ, 2019). Yılmaz and Kılıç have determined the albedo factor parameters of some marble samples. The chemical composition of selected samples was investigated by using neutron

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activation analysis (NAA). They reported that the shielding capacity of pink marble bigger than other marble samples. Also, the albedo factor parameter of pink marble reported that it is lower than the others (YILMAZ and KILIÇ, 2019). Yılmaz et al. have measured the reflection coefficients of Se, $H_3BO_3$, $NH_3–BH_3$, and $BH_3C_2H_5$ doped $VO_x$ thin films. Their experimental setup includes 5Ci $^{241}$Am radioactive sources that emit 59.54 keV gamma photons and HPGe semiconductor detector. They reported that the albedo factor parameters of $V_2O_5$thin films strongly changed by boron compound doping (YILMAZ et al., 2020). Ljubenov et al. have investigated the validation of expressing the possibility for reflection coefficients of low energy photons. For this purpose, they were used in different shielding materials. The experimental analysis was carried out by using photon reflection simulation codes MCNP, FOTELP, and PENELOPE of the Monte Carlo simulation program. They reported that it has been found that only the scaled albedo coefficients based on the average number of photon scatterings have a form of universal functions and these functions are determined by applying the least square method (LJUBENOV et al., 2010). Sabharwal et al. have measured the backscattering intensity distributions of 1.12 MeV gamma photons. The backscattering gamma rays have emitted from some selected target elements and alloy samples. They reported that in the same thicknesses the albedo factors (number, energy, and dose) and multiply scattering events have saturated (SABHARWAL et al., 2009). Kadotani and Shimizu have used the invariant embedding method in the measurement of the albedo factor parameters of selected target materials. The albedo parameters have measured depending on the angle and gamma-ray photon energy. They reported that the calculated albedos of selected samples were compared with Monte Carlo simulation program and closely agree with each other (KADOTANI and SHIMIZU, 1998). Kiran et al. have measured the albedo factor variation of aluminum, iron, and copper target samples as a function of incident gamma-ray photon energy and atomic number (Z). The gamma-ray photons are emitted from $^{57}$Co, $^{133}$Ba, $^{22}$Na, $^{137}$Cs, and $^{65}$Zn radioactive sources. The experimental measurements were compared with theoretical calculations that have gained by using the Monte Carlo simulation (KIRAN et al., 2016).

3d transition metals play an important role in the development of modern technology and knowing their valence electron structures is crucial for understanding their physical properties. Although some researchers have been made on the electronic structures of the 3d transition elements, no systematic and detailed study has been found to understand the valence electron structure of all these elements. Due to their wide range of physical properties, 3d transition metals have a very common use, as well as their compounds and alloys. Prinz has reported an article in Science journal that the $Fe_xNi_{1-x}$ alloys emerged as technologically important materials (PRINZ, 1998). Also, $Fe_xNi_{1-x}$ alloys are used in magnetic recorders, sensors, and magnetic memory cards due to their different properties such as low resistance and high permeability (BASA et al., 2002).

Lead and its different compounds were used in radiation applications as an absorber. Besides this important feature, the lead and its various compound has toxic effect in human health. This situation has been led new shielding material research (AKMAN et al., 2019b). In the present work, it is aimed to investigate the shielding properties of 3d alloy samples.3d transition elements are of great scientific and technological importance due to their properties originating from their electrons in their last orbital. Alloys made from these are used in electronic circuits in temperature control, magnetic recording and memory devices, etc. they are used in many applications. This present work is adapted to the determination of the albedo factor parameters of some 3d alloy targets and information about the samples was shown in Table 1. Besides, the experimental setup is including a 5Ci strength $^{241}$Am radioactive source and an HPGe semiconductor detector. The radioactive source and target materials were positioned according to the scattering experiment.

**Table 1**
The alloy samples used in the experimental study.

| Sample | | Mass (g) | Mass Thickness (g/cm$^2$) |
|---|---|---|---|
| **Alloy-1** | $Fe_{0.5}Cr_{0.5}$ | 0.295 | 0.222 |
| **Alloy-2** | $Fe_{0.7}Cr_{0.3}$ | 0.342 | 0.258 |
| **Alloy-3** | $Fe_{0.8}Cr_{0.2}$ | 0.268 | 0.203 |
| **Alloy-4** | $Fe_{0.9}Cr_{0.1}$ | 0.299 | 0.225 |
| **Alloy-5** | $Fe_{0.2}Ni_{0.8}$ | 0.253 | 0.191 |
| **Alloy-6** | $Fe_{0.3}Ni_{0.7}$ | 0.283 | 0.213 |
| **Alloy-7** | $Fe_{0.5}Ni_{0.5}$ | 0.240 | 0.181 |
| **Alloy-8** | $Fe_{0.6}Ni_{0.4}$ | 0.241 | 0.181 |
| **Alloy-9** | $Fe_{0.7}Ni_{0.3}$ | 0.228 | 0.172 |
| **Alloy-10** | $Fe_{0.8}Ni_{0.2}$ | 0.237 | 0.179 |

## 2. Experimental details and calculation methods

### 2.1. Experimental arrangement

The experimental setup and sample chamber were shown in Fig. 1 and Fig. 2, respectively. The systematic uncertainties were minimized or eliminated by using the sample chamber. Also, the sample chamber protects against the unfavorable effects of gamma-ray sources and the lead-coated sample chamber provides collimation to radioactive sources. In this way, the collimated gamma-ray beam has reached the center of targets. The alloy samples used in the experiment were obtained through medical companies. The semiconductor HPGe detector was connected to the computer-controlled system. The primary signals are transmitted to a multichannel analyzer firstly and finally reached the pc-based Genie-2000 program. The characteristic spectrum peaks are created from this program and the related spectrum areas of peaks calculated by using Origin 7.5 software.

### 2.2. Calculation of albedo factors

Albedo number, energy, and dose are known as albedo factors, in literature. The albedo factor parameters are calculating by using scattering information of the target and the radioactive source. Such as the ratio of reflected photons from target to the incident photons is called albedo number. Also, the scattering photon energy ratio to the incident photon energy is directly proportioned with the albedo energy. The last one of the albedo factor parameters is the albedo dose and calculated by using the albedo number and energy. The albedo dose of the selected sample is calculated via the sample and the air absorption coefficient ratios. The air consists of different elements in different percentages and the gaseous composition of the dry air is shown in Table 2.

The albedo number of known specimens is calculated via the following equation,

$$A_N = \left[\frac{N_{bs}/\varepsilon(E_{bs})}{(N_i/\varepsilon(E_i))(1/d\Omega)(1/2)}\right] \quad (1)$$

$N_{bs}$ and $N_i$represent the backscattered spectrum peak area and the incident spectrum peak area, respectively (SABHARWAL et al., 2011). $\varepsilon(E_{bs})$ and $\varepsilon(E_i)$ represents the photo-peak efficiencies of backscattered and incident energies corresponding to the detector, respectively. The photo-peak efficiency of the detector was determined by using different targets in the $13 \le Z \le 82$ range. Also, the target thicknesses are ranging from 0.0351 g/cm2 to 0.576 g/cm2. The solid angle of the detection system was calculated and represented by dΩ symbol. Only half of the gamma photons emitted by the radioactive source are incident on the target. Because of this situation, the 1/2 factor is used in the equation.

Coherent to Compton scattering intensity ratio was plotted as a function of the atomic number and given by Fig. 3. The related figure is a kind of calibration curve and derived from the Compton to coherent scattering ratios of pure targets in the atomic range $4 \le Z \le 82$. According to this calibration curve the atomic number of alloy samples were derived by using their scattering ratios. Also, the ratio of the incident

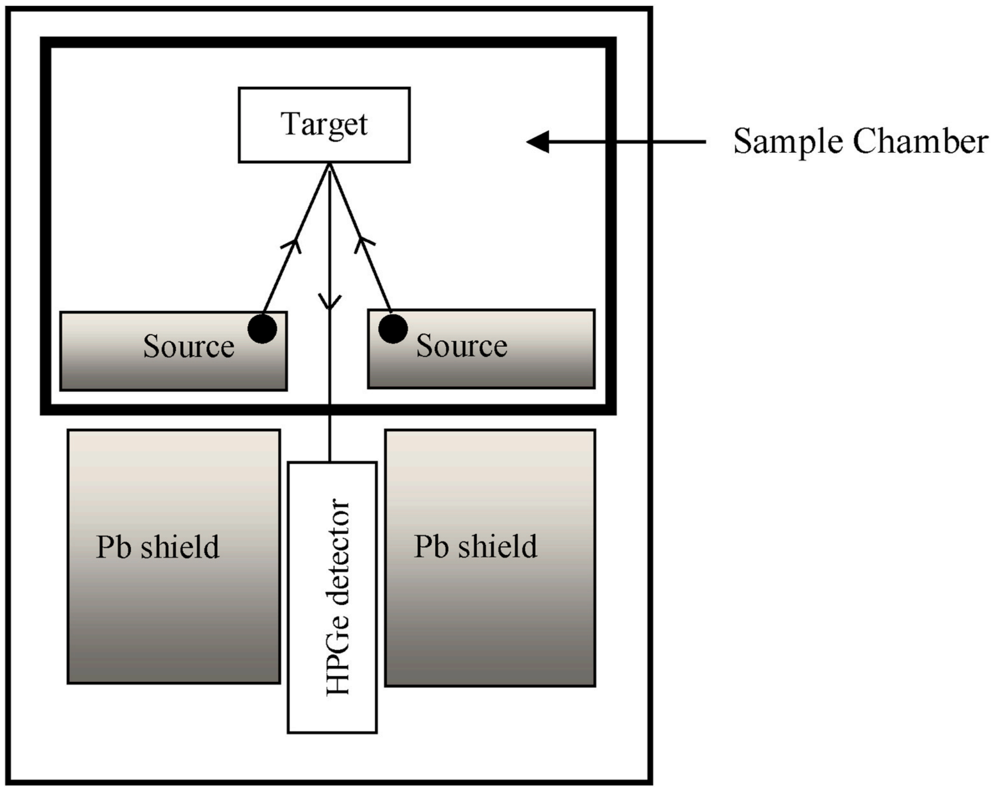


**Fig. 1.** Experimental setup (radius of collimator is 0.53 cm). This figure is only a schematic diagram of experimental setup.

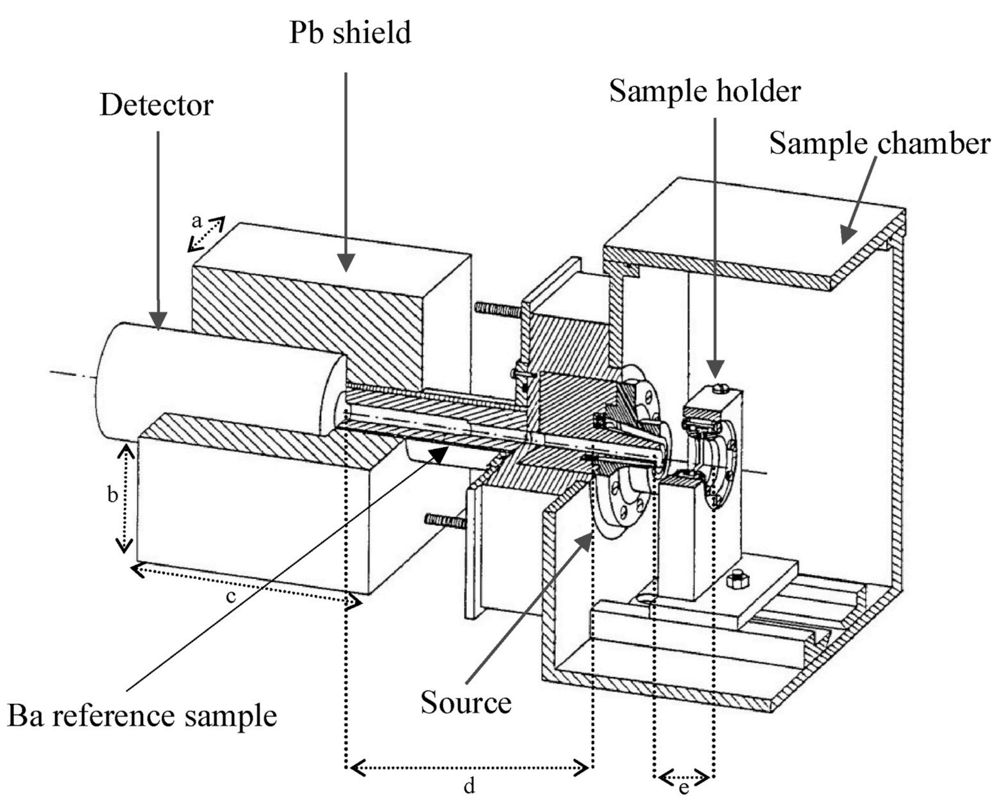


**Fig. 2.** Sample chamber (a = 6.5 cm, b = 6.3 cm, c = 13.5 cm, d = 11 cm, e = 5 cm).

**Table 2**
Table of gaseous composition of dry air.

| Constituent | Chemical symbol | Mole percent |
|---|---|---|
| Nitrogen | $N_2$ | 78.084 |
| Oxygen | $O_2$ | 20.947 |
| Argon | Ar | 0.934 |
| Carbon dioxide | $CO_2$ | 0.0350 |
| Neon | Ne | 0.001818 |
| Helium | He | 0.000524 |
| Methane | $CH_4$ | 0.00017 |
| Krypton | Kr | 0.000114 |
| Hydrogen | $H_2$ | 0.000053 |
| Nitrous oxide | $N_2O$ | 0.000031 |
| Xenon | Xe | 0.0000087 |
| Ozone | $O_3$ | 0.00000001 |

photon energy to the backscattered photon energy is known as albedo energy and calculated by the following equation,

$$A_E = [\frac{E_{bs}}{E_i}]A_N \tag{2}$$

the backscattered and incident photon energies represent by $E_{bs}$ and $E_i$, respectively in the equation. The following equation represents the equation of the albedo dose calculations

$$A_D = [\frac{\sigma_a(E_{bs})}{\sigma_a(E_s)}]A_E \tag{3}$$

the albedo dose is proportional to the albedo energy and, represents the energy absorption coefficient of air for the backscattered and the incident gamma photons, respectively. The air is composed of different integrant materials in different combinations of percentages. The chemical percentages of the air are taken from Table 1. Also, the backscattered and incident gamma energies have determined from the characteristic spectrum and the absorption coefficients of related spectrums were taken from the WinXCOM photon cross-section database (GERWARD et al., 2001).

## 3. Result and discussion

In nature, we do not have the chance to conduct a radiation absorption dose or the radiation sensitivity test for all living or non-living materials present in our habitat. Therefore, it makes more sense to make general inferences with some material. Within the framework of this logic, we can reach the general opinion if the properties of some scientifically important materials for radiation shielding are determined. Also, there are many recent studies on this subject. The radiation absorption coefficient is the most commonly used parameter that indicates the degree of shielding. In addition to the absorption coefficient, the albedo factor parameters are also used to determine the degree of radiation protection of materials.

The albedo factor calculation and determination of 3d alloy samples and a brief explanation of the experimental arrangement is given this section. The experimentally calculation and determination of 3d alloy target samples were carried out by using 5Ci strength $^{241}$Am radioactive sources that is emitting 59.54 keV energy gamma-ray photons in 160°scattering angle. The experimental arrangement has a lead-coated conical source collimator and this component prevent unwanted counts from reaching the detector. Related unwanted count collimation system has presented in the sample chamber. Lack of the sample chamber shielding is led to unwanted counting and this situation has causes background counting. Although all precautions were considered, the systematic counting originating from the background was subtracted from characteristic spectral counting.

3d transition alloy specimens were used as a target in experimental applications. The selected alloy samples have scientific importance because of their wide usage at electronic circuits in temperature control, magnetic recording and memory devices, and many applications. Fig. 4 represents the typically recorded spectrum at 59.54 keV for selected alloy samples. Besides, the albedo factor parameter values of 3d alloy samples were plotted as a function of atomic number and shown in Figs. 5–7, respectively. According to the related figures, the albedo factor parameters (albedo number, energy, and dose) were exponentially decreased with increasing atomic numbers. The experimental

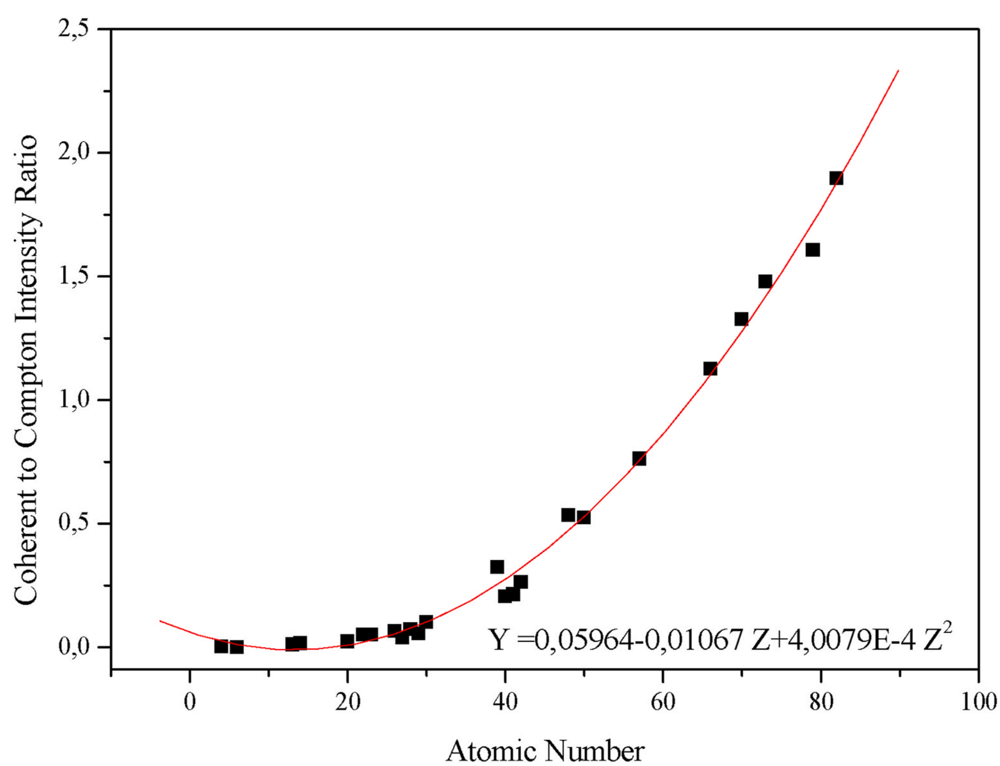


**Fig. 3.** Coherent to Compton scattered intensity as a function of atomic number.

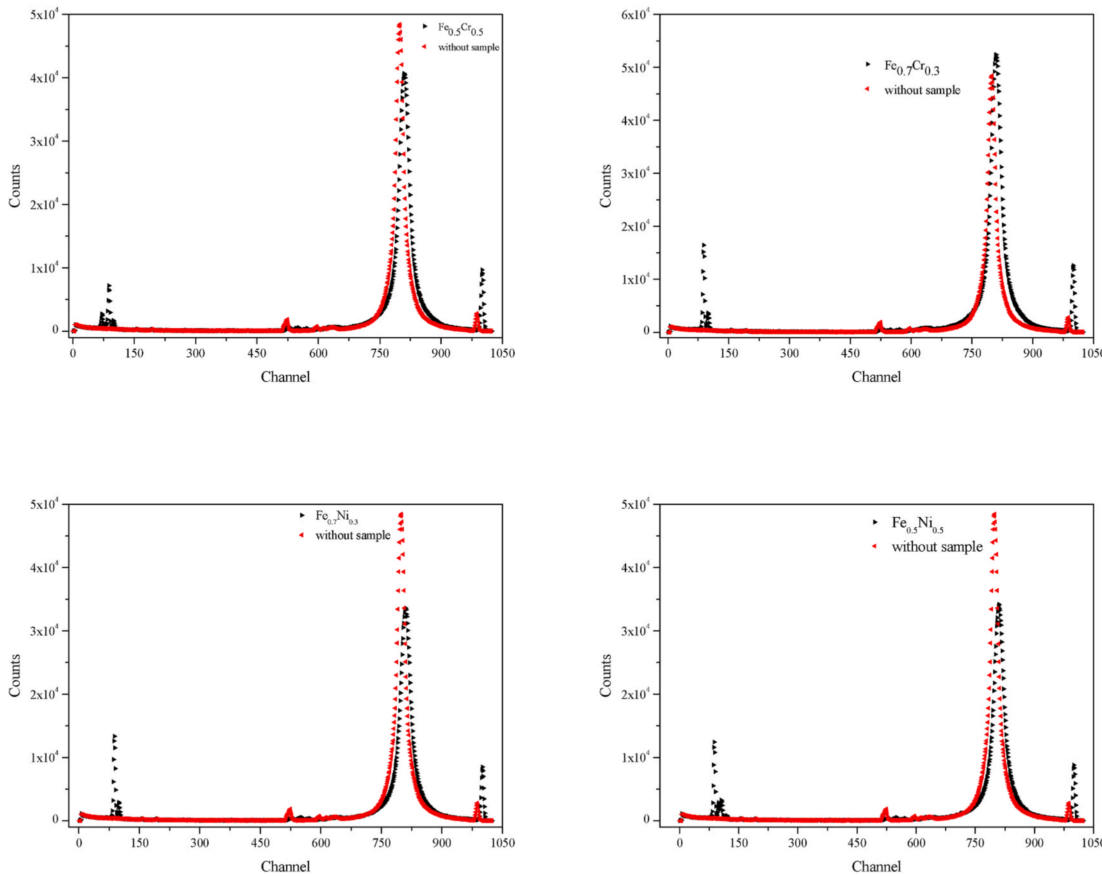


**Fig. 4.** Typical spectrum of the counting system with alloy samples and without alloy samples.

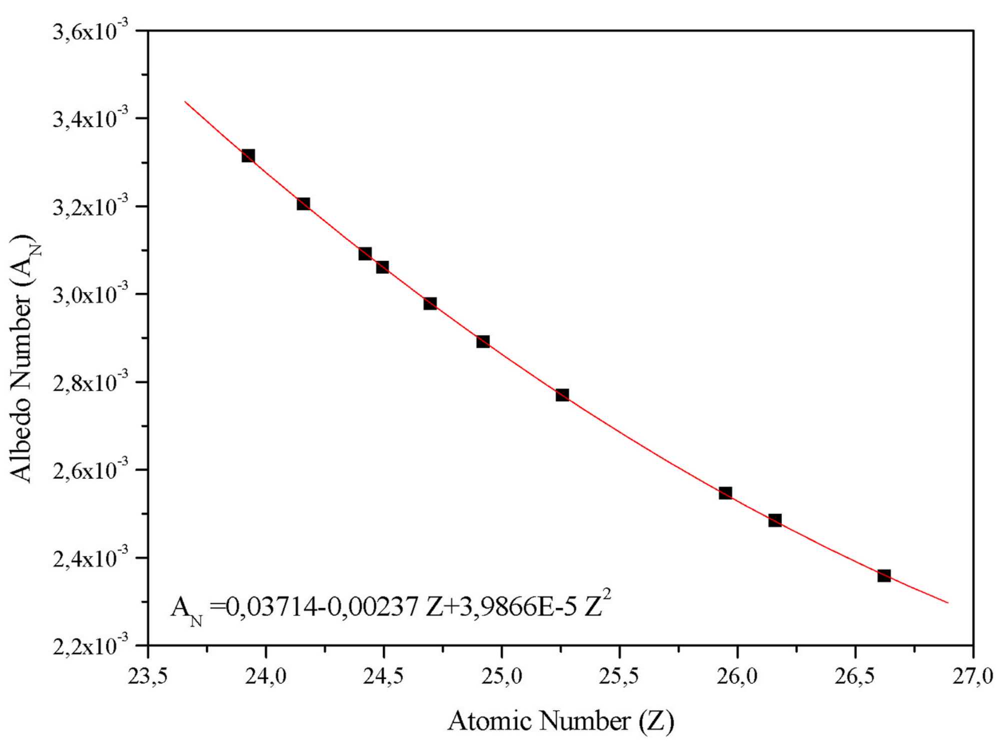


**Fig. 5.** The albedo number distribution of alloy samples as a function of the atomic number.

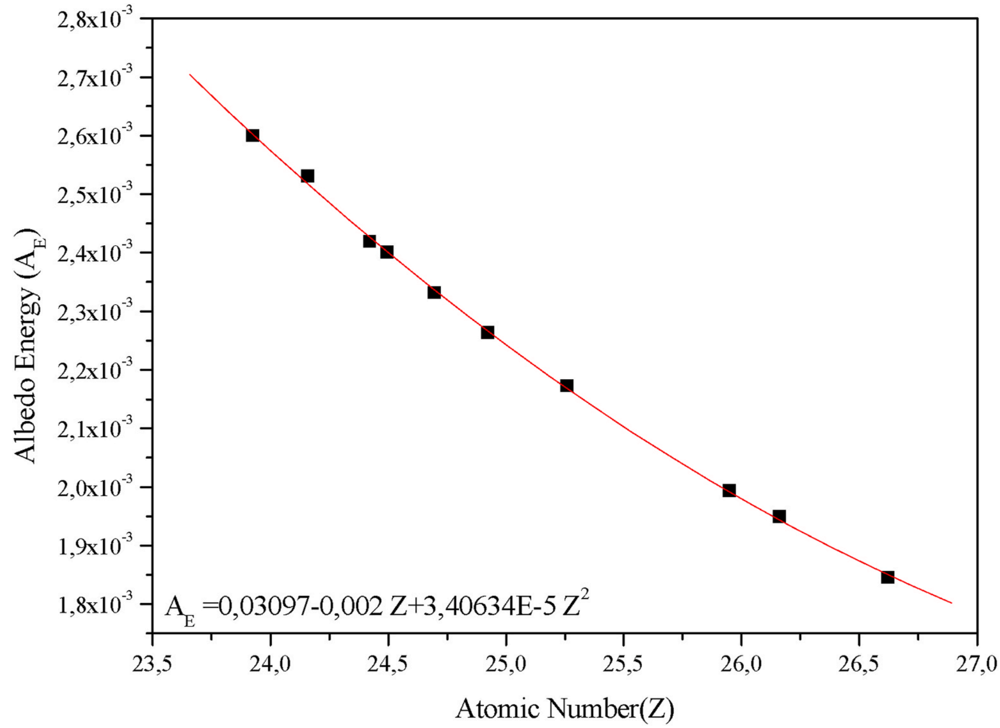


**Fig. 6.** The albedo energy distribution of alloy samples as a function of the atomic number.

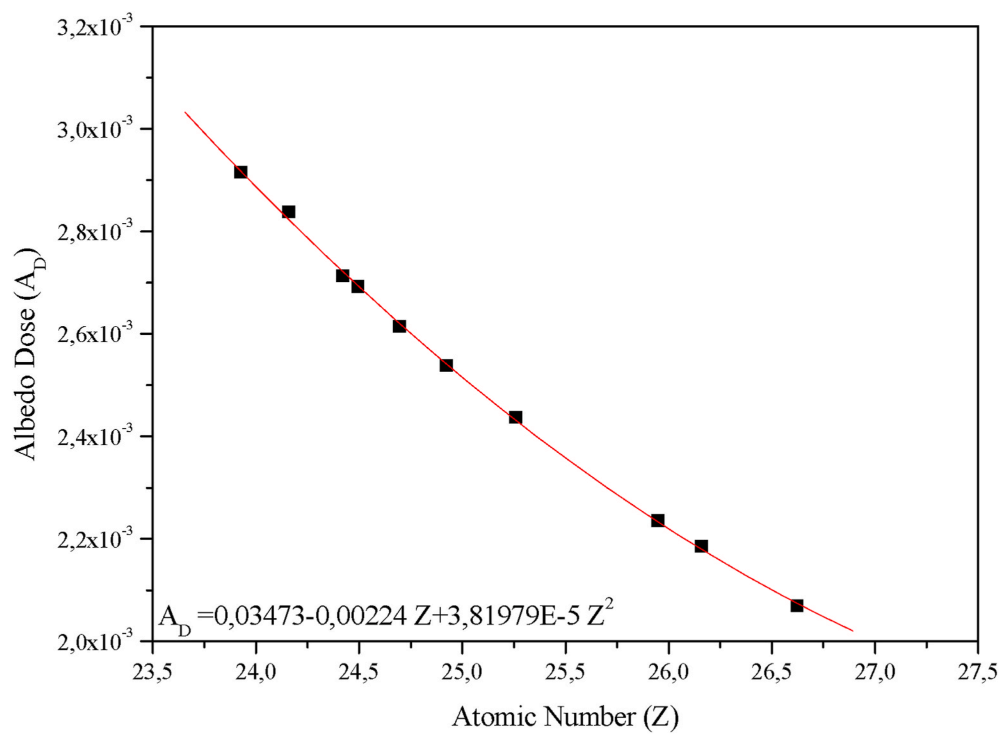


**Fig. 7.** The albedo dose distribution of alloy samples as a function of the atomic number.

application was performed at low energy region. So, the backscattering process cross-section is more dominant than other scattering process cross-sections in this region. The Compton and coherent scatterings are parts of the backscattering processes. Also, the scattering intensity ratio information of these events is used in the albedo calculations.

3d alloy samples have valence electrons on their outer shells and multiple scattering can occur on these free electrons. Also, the backscattering on the stagnant or single electrons of target samples is called the Compton scattering, as we know. Besides, these free valence electrons are causes tailing on low and high energy regions of the Compton scattering spectrum peak. Against the broadening on the low and high energy region of the Compton scattering, the reverse Compton scattering effect leads to strengthening in the coherent spectrum peaks. Besides, the atomic number of used targets and the saturation capability are proportional to each other. So, the increasing atomic number of the target samples has raised the saturation of incident gamma-rays. The experimental measurements show that the theoretical calculations are in good agreement with each other (YILMAZ et al., 2017).

## 4. Conclusion

The albedo factor parameters of samples are related to the atomic number of the target specimens. The photoelectric cross-section of the samples is increasing with the increasing atomic mean atomic number in compounds and, atomic numbers in elements. Also, the increasing atomic number leading decreases in the Compton scattering cross-section of samples. This present work is indicating the increments of the atomic number of target samples that were prepared from 3d transition metals are causing decreasing in the albedo factor parameters. Also, the scattering center shifting, and the effect of the reverse Compton scattering was caused by the albedo factor parameter decreasing of targets. The mentioned albedo factor parameter decreasing was taken place in the same scattering angle and incident photon energies.

The albedo factor parameter calculations offer a simple alternative to the determination of the backscattering capabilities of target samples (BROCKHOFF and Carl, 2003). Therefore, the albedo factor investigation of different alloy samples and different compositions of compounds have attracted the researchers. Although much investigation has done via different experimental studies on this area, new experimental techniques and different compounds are required for technological innovations. These required studies could be carried out by using different gamma energies, different samples, and different thicknesses of the same samples.

## CRediT authorship contribution statement

**Turşucu Ahmet:** Data curation, Writing - original draft, Conceptualization, Methodology, Software, Writing - review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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